Smoothing and first-counterexample scalar gates

27 October 2026

Scope. This note independently checks the scalar step of a proposed proof redesign. It does not establish the acquisition or local-exclusion theorems, verify their certificates, or authorize removal of a proof component. Baseline: `6346fc1b4432bc5982822aa696eb566ed9e1e1ee` in the fork `kolodziejek/berry-esseen-certificates`.

Let $\mathcal{P}$ be all real laws with mean zero, variance one and finite $\rho(P) = \mathbb{E}_P|X|^3$. For iid X_i with law P, put

$$S_n = \sum_{i=1}^{n} X_i, \qquad \Delta_n(P) = \sup_{H\in\mathbb{R}} |\mathbb{P}_P(S_n \le H) - \Phi(H/\sqrt{n})|.$$

$$C_n = \sup_{P\in\mathcal{P}} \frac{\sqrt{n}\,\Delta_n(P)}{\rho(P)}, \qquad C_{\mathrm{E}} = \frac{3+\sqrt{10}}{6\sqrt{2\pi}}.$$

Thus H always denotes a raw sum threshold. All terminating decimals below denote exact rationals. Lyapunov's inequality gives $\rho(P) \ge 1$.

Lemma 1 (Support-free smoothing)**.** *For every integer $n \ge 2$,*

$$C_n \le \sqrt{\frac{n}{n-1}}\, C_{n-1} + \frac{\sqrt{n}}{15(n-1)^{3/2}} + \frac{2\sqrt{n}}{25(n-1)^2}. \tag{1}$$

No bounded support, extremizer or higher moment is assumed.

Proof. Fix $P \in \mathcal{P}$ and $H \in \mathbb{R}$, write $v = n-1 \ge 1$, and set $F_v(t) = \Phi(t/\sqrt{v})$. Conditioning on X_n and applying the defining bound for C_v to the same marginal at every threshold gives

$$|\mathbb{P}_P(S_n \le H) - \mathbb{E}F_v(H - X_n)| \le C_v\rho/\sqrt{v}. \tag{2}$$

Only the smooth Gaussian function is expanded. Taylor's theorem gives

$$|F_v(H-y) - F_v(H) + yF_v'(H) - \tfrac{1}{2}y^2F_v''(H)| \le \frac{\|\phi''\|_\infty}{6v^{3/2}}|y|^3.$$

In fact $\|\phi''\|_\infty = \phi(0)$: since $\phi''(z) = (z^2-1)\phi(z)$, the assertion is immediate on $z^2 \le 1$; on $u = z^2 \ge 1$, $(u-1)e^{-u/2}$ is maximized at $u = 3$, with value $2e^{-3/2} < 1$. Taking expectation therefore yields

$$|\mathbb{E}F_v(H - X_n) - F_v(H) - \tfrac{1}{2}F_v''(H)| \le \frac{\phi(0)\rho}{6v^{3/2}} < \frac{\rho}{15v^{3/2}}. \tag{3}$$

The elementary bound $\phi(0) < 2/5$ is established below.

Direct differentiation in the variance gives $\partial_v F_v = F_v''/2$ and $\partial_v^2 F_v = F_v^{(4)}/4$, where primes differentiate the raw argument. Hence

$$F_{v+1}(H) - F_v(H) - \tfrac{1}{2}F_v''(H) = \frac{1}{4}\int_0^1 (1-t)F_{v+t}^{(4)}(H)\,dt.$$

Since $F_w^{(4)}(H) = w^{-2}\phi'''(H/\sqrt{w})$, its absolute value is at most $\|\phi'''\|_\infty/(8v^2)$. The elementary Gaussian Fourier identity, differentiated under its absolutely convergent integral, implies

$$\|\phi'''\|_\infty \le \frac{1}{2\pi}\int_{\mathbb{R}} |t|^3 e^{-t^2/2}\,dt = \frac{2}{\pi}.$$

Indeed the positive-half-line integral is 2, by $u = t^2/2$. The heat remainder is consequently at most $1/(4\pi v^2) < 2/(25v^2)$. Combine this with (2)–(3), multiply by $\sqrt{n}/\rho$, and use $\rho \geq 1$. The resulting estimate is uniform in H and P, so both suprema are legitimate.

For a strict event, let $H_j \uparrow H$: its probability is the limit of the closed-event probabilities, while Φ is continuous. Thus the same bound covers both jump edges and reflection. Neither an indicator nor the distribution function of S_v was differentiated. Rare remote atoms need only the assumed third absolute moment. □

Lemma 2 (Elementary constants)**.** *One has* $\pi > 157/50 > 25/8$ *and* $0 < C_{\mathrm{E}} < 41/100$.

Proof. Let $a = \arctan(1/5)$ and $b = \arctan(1/239)$. Tangent addition gives $\tan(2a) = 5/12$, $\tan(4a) = 120/119$, and $\tan(4a-b) = 1$. Integrating $1-t^2 < 1/(1+t^2) < 1$ on positive intervals gives

$$0 < 4(1/5 - 1/(3 \cdot 5^3)) - 1/239 < 4a - b < 4/5 < 1 < \pi/2.$$

Here $\pi/2 > 1$ follows from $\arctan(1) > 1/2$. Thus $4a - b = \pi/4$ and

$$\pi > 16(1/5 - 1/(3 \cdot 5^3)) - 4/239 = \frac{281476}{89625} > \frac{157}{50} > \frac{25}{8}.$$

This also proves $\phi(0) < 2/5$ and $1/(4\pi) < 2/25$. Since $\sqrt{10} < 19/6$,

$$C_{\mathrm{E}}^2 = \frac{19 + 6\sqrt{10}}{72\pi} < \frac{1900}{11304} < \frac{1681}{10000}.$$

The last difference is $2024/113040000 > 0$. □

Lemma 3 (Scalar entry from sharp predecessors)**.** *Put* $c = C_{\mathrm{E}}$. *If* $n \geq 47$, $C_{n-1}, C_{n-2} \leq c$ *and* $C_n \geq c$, *then*

$$d_n := n(C_{n-1} - C_n)_+ = 0, \qquad 0 \leq n(C_n - c) < \frac{335909}{1150000} < \frac{3}{10},$$

and

$$C_n < .4247, \qquad C_{n-1}, C_{n-2} < .4253.$$

If $C_n > c$, *the lower bound on the excess is strict.*

Proof. Set $s = \sqrt{n/(n-1)} > 1$. From (1) and $s - 1 = 1/((n-1)(s+1))$,

$$n(C_n - c) \leq \frac{n}{n-1}\left(\frac{c}{s+1} + \frac{s}{15} + \frac{2s}{25\sqrt{n-1}}\right).$$

For all $n \geq 47$, $n/(n-1) \leq 47/46$, $\sqrt{n-1} > 6$ and $s < 1011/1000$; the last assertion follows from $46 \cdot 1011^2 - 47 \cdot 1000^2 = 17566 > 0$. Hence

$$n(C_n - c) < \frac{47}{46}\left(\frac{41}{200} + \frac{1011}{15000} + \frac{1011}{75000}\right) = \frac{335909}{1150000} < \frac{3}{10}.$$

The margin below $3/10$ is $9091/1150000$. Also

$$C_n < \frac{41}{100} + \frac{3}{470} = \frac{1957}{4700} < \frac{4247}{10000}, \qquad C_{n-j} \leq c < \frac{41}{100} < \frac{4253}{10000} \quad (j = 1, 2).$$

The own-index-cap margin is $3909/470000$. The predecessor sign gives $d_n = 0$. Every comparison includes the endpoint $n = 47$. □

Conditional exclusion interface. For the next two propositions, an exclusion input includes: attainment of an oriented positive closed extremum; acquisition of all required same-law support, probability, moment and resource bounds; and the complete local exclusion, with every sign, normalization, parameter-domain and numerical-certificate hypothesis discharged. Scalar entry alone does not establish this interface.

Proposition 4 (Weak global assembly)**.** *Assume $C_j \le c = C_{\mathrm{E}}$ for $1 \le j \le 46$. Suppose the exclusion interface excludes every selected owner at an index $n \ge 47$ satisfying*

$$0 < C_n - c \le .7/n, \quad d_n \le .19, \quad C_n < .4247, \quad C_{n-1}, C_{n-2} < .4253.$$

Then $C_n \le c$ for every positive integer n.

Proof. If some $C_n > c$, choose the least such n. Then $n \ge 47$ and both predecessors are at most c. Lemma 3 supplies all the scalar gates, contradicting the stipulated exclusion. □

Proposition 5 (Separately named equality-inclusive strict tail)**.** *Assume the same weak prefix through 46. Suppose the exclusion interface instead applies also at equality, replacing $0 < C_n - c$ above by $0 \le C_n - c$, and excludes those owners as well. Then $C_n < c$ for every $n \ge 47$.*

Proof. If the set $\{n \ge 47 : C_n \ge c\}$ is nonempty, take its least member. All earlier tail values are $< c$; prefix values are $\le c$. Both predecessors are therefore at most c. Lemma 3 again supplies the scalar gates, including when $C_n = c$, contradicting the equality-inclusive exclusion. □

The two propositions have different hypotheses and contrary sets. Proposition 4 does not imply strictness: the constant sequence $C_n = c$ itself satisfies the scalar recurrence. For Proposition 5, the least $\ge c$ failure is chosen only within the tail; equality inside the weak prefix is permitted.

Source and verification boundary. At the frozen baseline, the recurrence is `paper/analytic-supplement.txt`, physical lines 6479–6522 (label `cut49:global:smoothing`, original M7458–7506). The existing tail assembly is `paper/main.tex`, lines 460–510. The supplied attainment proof expressly includes equality at lines 107–136; supplement lines 719–755 also permit zero excess in a support refinement. These observations do not establish equality compatibility of every intervening acquisition edge. The local comparison and its binomial input are at supplement lines 7153–7197 and 7748–7768.

The accompanying `check_scalar.py` uses only exact integers and rational arithmetic to check the finite comparisons. Its fresh run `runs/least-counterexample-20261027/scalar01/` returned zero; all thirteen negative controls rejected. That arithmetic PASS proves neither the analytic steps nor the exclusion interface. This note identifies a conditional replacement for scalar-entry computations, not a certificate-deletion list.